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United States Patent Application Publication

20250286314

Kind Code

A1

Publication Date

September 11, 2025

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System for Connecting Two Separate Busbars to a Coaxial Busbar

Abstract

The system for connecting two separate busbars to a coaxial busbar optimizes the electrical connection in electromobility and energy storage systems. A housing assembly integrates specific receptacles for coaxial busbars and separate busbars, supported by intermediate busbars that are connectable in an electrically conductive manner, in particular connectable in a cohesive bond manner. The connection assembly facilitates efficient and safe assembly and provides flexibility for various applications. The assembly provides a robust and adaptable solution for the integration in compact as well as power-intensive systems, making it an attractive choice for advanced applications.

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Appl. No.: 19/073102

Filed: March 07, 2025

Foreign Application Priority Data

DE 102024106867.1

Mar. 11, 2024

Publication Classification

Int. Cl.: H01R13/621 (20060101); H01R13/447 (20060101); H01R13/506 (20060101);
H01R25/16 (20060101)

U.S. Cl.:

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to and the benefit of German Patent Application No. 102024106867.1 filed Mar. 11, 2024, which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

[0002] The invention relates to a system for connecting busbars.

[0003] Busbars are used in the field of electromobility to carry the comparatively high currents for charging the batteries or for providing the drive power in electric vehicles. Different systems are used for connecting the busbars.

[0004] Since the installation space in electric vehicles is often very limited, systems for connecting busbars are required that are as compact as possible, that at the same time transmit the high currents with low losses, and are easy to manage.

[0005] The object of the present invention is therefore to create such a system.

SUMMARY OF THE INVENTION

[0006] According to the invention, the above-mentioned object is satisfied by a system for connecting two separate busbars to a coaxial busbar, which comprises at least two busbars arranged coaxially to one another, with a housing assembly which comprises a receptacle for the coaxial busbar, wherein the receptacle extends in an direction of insertion, as well as a first receptacle for the one of the two separate busbars and a second receptacle for the other of the two separate busbars, with a first intermediate busbar for connecting the receptacle for the coaxial busbar to the first receptacle, wherein the first intermediate busbar is configured to be connectable to the one busbar of the coaxial busbar in an electrically conductive manner and to a second intermediate busbar for connecting the receptacle for the coaxial busbar to the second receptacle, wherein the second intermediate busbar is configured to be connectable to the other busbar of the coaxial busbar in an electrically conductive manner, and wherein the first and the second intermediate busbar can be arranged in the receptacle in the direction of insertion spaced apart from one another.

[0007] The direction of insertion is the direction in which the coaxial busbar can be inserted into the housing assembly. This direction can correspond to the longitudinal extension of the coaxial busbar and enables precise positioning within the housing assembly.

[0008] A high degree of reliability and safety is obtained by introducing optimizations in the mechanical configuration and electrical performance of the system according to the invention. This includes not only the mechanical integrity and electrical conductivity, but also aspects such as touch protection and thermal management capability which are required for long-term and safe operation under various operating conditions.

[0009] The above solution can be improved by the following independent, individually advantageous, and freely combinable further developments.

[0010] In a preferred embodiment of the system according to the invention, the current-conducting connection between the first intermediate busbar and the one busbar of the coaxial busbar and/or the current-conducting connection between the second intermediate busbar and the other busbar of the coaxial busbar is or are configured as a cohesive bond connection.

[0011] A further preferred embodiment of the system according to the invention is equipped with a first touch-protected fastening element that can be operated from outside the housing assembly and that projects through the first intermediate busbar into the first receptacle, and with a second touch-protected fastening element that can be operated from outside the housing assembly and that

projects through the second intermediate busbar into the second receptacle.

[0012] In particular, according to a further particularly preferred embodiment of the invention, it is provided that at least one of the fastening elements is configured as a touch-protected screw.

[0013] A particularly advantageous further development of the system according to the invention provides that the system comprises at least one threaded sleeve which is fastened to one of the two separate busbars, in particular inserted into an opening in the latter, and screwed to a touch-protected screw.

[0014] The threaded sleeves are preferably made of steel in order to withstand higher mechanical loads.

[0015] In particular, a wall of the housing assembly can be disposed between the first and the second receptacle. This can prevent, for example, creepage currents and improve the electrical insulation of the two busbars.

[0016] It can also be provided that the first screw ends in the first receptacle and the second screw in the second receptacle.

[0017] The intermediate busbars can preferably each be made of copper. Aluminum or another material with a comparably low electrical resistance is also possible.

[0018] The touch-protected screw can be configured such that its screw head and/or oppositely disposed screw end are overmolded with plastic, wherein a screw shaft made of steel a shoulder with an enlarged diameter that rests on a surface of the intermediate busbar facing away from the respective busbar of the two separate busbars.

[0019] The threaded sleeve can be fitted into an opening in the respective busbar of the two separate busbars and can abut with a radially outwardly projecting collar against a surface of the respective busbar of the two separate busbars facing away from the respective intermediate busbar.

[0020] The opening provided in the respective busbar of the two separate busbars can be configured as a round hole that is closed all the way around or open on one side, or as an elongate hole that is closed all the way around or open on one side.

[0021] In a further preferred embodiment of the system according to the invention, the first receptacle and the second receptacle extend in the housing assembly in a direction that runs transversely to the direction of insertion.

[0022] In a further preferred embodiment of the system according to the invention, it is provided that the first intermediate busbar and the second intermediate busbar each comprise a first connecting section which is arranged in the receptacle for the coaxial busbar, and a second connecting section which is arranged either in the first receptacle for the one of the two separate busbars or in the second receptacle for the other of the two separate busbars.

[0023] In yet another particularly advantageous embodiment of the system according to the invention, it is provided that the first connecting section is configured to be sleeve-shaped or hollow-cylindrical at least in part or planar at least in part.

[0024] According to yet another preferred embodiment of the invention, it is provided that the main direction of extension of the first connecting section is changed as compared to a main direction of extension of the first connecting section, wherein this change in direction is formed in particular by an angular deformation between the connecting sections.

[0025] A further advantageous embodiment of the system according to the invention provides that the second connecting section be configured as a fastening tab and this fastening tab comprises an opening through which the corresponding fastening element is guided.

[0026] The opening provided in the fastening tab can be configured as a round hole that is closed all the way round or open on one side or as an elongated hole that is closed all the way around or open on one side.

[0027] In a system according to the invention according to a further preferred embodiment of the invention, the opening in the fastening tab is insulated from the respective busbar of the two separate busbars.

[0028] For example, the first screw and the second screw can each be guided through a plastic sleeve that is inserted into openings in the first and the second intermediate busbar and extends from these openings to the associated first and second threaded sleeve.

[0029] The intermediate busbar and the busbar of the two separate busbars connected thereto can be directly in contact with one another. Alternatively, the intermediate busbar and the connected busbar of the two separate busbars can be spaced apart from one another, and an electrical connection between these busbars can be established exclusively by way of the screw connection.

[0030] Another particularly expedient embodiment of the system according to the invention provides that the first intermediate busbar and the second intermediate busbar each be configured as L-shaped elements.

[0031] A particularly advantageous development of the system according to the invention provides that the housing assembly comprises a first housing half in which the receptacle for the coaxial busbar is disposed and a second housing half that can be plugged together with the first housing half and in which both the first receptacle as well as the second receptacle are disposed.

[0032] In particular, the first housing half and the second housing half can each be made of plastic, in particular of glass fiber reinforced plastic.

[0033] According to a particularly advantageous embodiment of the system according to the invention, the intermediate busbars are arranged at least in part in the second half of the housing.

[0034] The intermediate busbars can preferably each be configured as multi-layer metal sheets that ensure flexibility perpendicular to the sheet plane. Multi-layer metal sheets improve adaptability due to their increased flexibility and optimize current conduction by reducing resistance losses and increasing electrical efficiency.

[0035] A further advantageous embodiment of the invention arises from the fastening elements being attached to the first housing half.

[0036] In particular, the screw head of the screws can each comprise a plastic external thread that is screwed into an internal thread of the second housing half, and the screw shaft of the screws can each comprise an external thread that is screwed into an internal thread of the threaded sleeve, which is preferably made of steel.

[0037] In particular, a fastening opening in the second housing half, through which the respective screw is inserted, can be covered at least in part by a screw holder that is attached in a form-fit manner to the second housing half in order to enable the screw to be secured.

[0038] Furthermore, according to the invention, a connection assembly is provided which comprises a system as described above, a coaxial busbar and two separate busbars, wherein one of the two separate busbars is connected to the first intermediate busbar by way of a first fastening element in the first receptacle and the other of the two separate busbars is connected to the second intermediate busbar by way of a second fastening element in the second receptacle.

[0039] The connection assembly according to the invention provides a number of advantages that make it particularly attractive for use with modern electromobility and energy system applications. By combining coaxial and intermediate busbars, the assembly improves electrical conductivity and minimizes losses, while the robust construction and touch-protected components ensure high mechanical stability and safety. The simple assembly enables quick and flexible installation that can easily be adapted to different requirements.

[0040] The at least two busbars of the coaxial busbar can be made of aluminum or an alloy containing aluminum.

[0041] In particular, the at least two busbars of the coaxial busbar can be separated from one another by an intermediate insulation and surrounded by an external insulation.

[0042] A preferred further development can provide that the coaxial busbar comprises exactly two busbars and is configured to be cylindrical, wherein the first busbar forms a cylindrical core and the second busbar forms a hollow cylindrical shell within the coaxial busbar.

[0043] The intermediate busbars can each comprise a hollow cylindrical section that encloses a

circumference of the respective busbar of the coaxial busbar and a planar section that protrudes from this respective busbar and comprises an opening for the screw to pass through.

[0044] An alternative preferred development can provide that the coaxial busbar comprises exactly two busbars and a flat rectangular cross-sectional contour, wherein the first busbar forms a flat, rectangular core in cross-section and the second busbar forms a flat, rectangular and hollow shell in cross-section within the coaxial busbar.

[0045] The intermediate busbars can each be configured as L-shaped angled elements in which one leg rests over the entire surface on a long underside of the respective busbar of the coaxial busbar and a leg angled away from it comprises an opening for passing the screw through.

[0046] The two separate busbars can be arranged parallel adjacent to one another, in particular parallel adjacent to one another offset in a direction that runs parallel to a direction of longitudinal extension of the coaxial busbar.

[0047] The two intermediate busbars can also be arranged parallel adjacent to one another, in particular parallel adjacent to one another offset in a direction that runs parallel to a direction of longitudinal extension of the coaxial busbar.

[0048] An advantageous embodiment of the connection assembly according to the invention provides that the fastening tabs of the intermediate busbars each directly abut a corresponding one of the two separate busbars.

[0049] According to a preferred embodiment of the invention, in the receptacle for the coaxial busbar, the intermediate busbars are each connected in a cohesive bond manner to a corresponding busbar of the coaxial busbar.

[0050] The cohesive bond connection between the first (second) intermediate busbar and the one (the other) of the two busbars of the coaxial busbar can be established by welding, in particular by torsional ultrasonic welding, friction welding, electron beam welding, nanofoil welding or roll seam welding, or by soldering, in particular by induction brazing.

[0051] The receptacle for the coaxial busbar can comprise an inner contour perpendicular to the direction of insertion which corresponds to the outer contour of the coaxial busbar together with the intermediate busbars attached thereto perpendicular to the longitudinal extension of the coaxial busbar. This allows the coaxial busbar provided with the intermediate busbars to be inserted in a simple manner into the receptacle for the coaxial busbar.

[0052] According to a further embodiment of the invention, it is further provided that the first housing half is configured to be pre-assembled with the fastening elements, the coaxial busbar and the intermediate busbars and the second housing half is configured such that it can be plugged together with the first housing half.

[0053] In a particularly preferred manner, the second housing half is configured to be pre-assembled with the two separate busbars.

[0054] A cooling device can be integrated into the construction of the connection assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0055] The above embodiments are explained in more detail hereafter by way of embodiments with reference to the drawings. The same reference numerals are used for elements that correspond to one another in terms of function and/or structure.

[0056] In accordance with the above, a feature of the embodiment described hereafter can be omitted if the technical effect associated with this feature is not important for a specific application. Conversely, in accordance with the above observations, a feature not present in the embodiment described hereafter can be added if the technical effect associated with this feature is important for a specific application, where:

[0057] FIG. 1 is a schematic perspective exploded view of a first embodiment with a coaxial busbar;

[0058] FIG. 2 is a schematic cross-sectional view of the embodiment of FIG. 1;

[0059] FIG. 3 is a schematic perspective exploded view of a second embodiment with a coaxial busbar;

[0060] FIG. 4 is a schematic cross-sectional view of the embodiment of FIG. 3;

[0061] FIG. 5 is a schematic perspective exploded view of a third embodiment;

[0062] FIG. 6 is a schematic cross-sectional view of the embodiment of FIG. 5;

[0063] FIG. 7 is a schematic perspective exploded view of a fourth embodiment;

[0064] FIG. 8 is a schematic cross-sectional view of the embodiment of FIG. 7;

[0065] FIG. 9 is a schematic perspective cross-sectional view of a fifth embodiment; and

[0066] FIGS. 10 to 12 are a schematic representation of an assembly sequence for producing a sixth embodiment.

DETAILED DESCRIPTION

[0067] The description of illustrative embodiments according to principles of the present invention is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments of the invention disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. In various applications, relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “top” and “bottom” as well as derivative thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation unless explicitly indicated as such. Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” and similar refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. Moreover, the features and benefits of the invention are illustrated by reference to the preferred embodiments. Accordingly, the invention expressly should not be limited to such preferred embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features, the scope of the invention being defined by the claims appended hereto.

[0068] Exemplary embodiments of the present invention are now described with reference to the Figures. Reference numerals are used throughout the detailed description to refer to the various elements and structures. Although the following detailed description contains many specifics for the purposes of illustration, a person of ordinary skill in the art will appreciate that many variations and alterations to the following details are within the scope of the invention. Accordingly, the following embodiments of the invention are set forth without any loss of generality to, and without imposing limitations upon, the claimed invention.

[0069] Systems 20 shown in the figures are used to connect a coaxial busbar 4 to two separate busbars 6a, 6b. This coaxial busbar 4 comprises two coaxially arranged busbars 4a, 4b which are typically made of aluminum or an aluminum alloy to ensure high electrical conductivity and mechanical stability. These two busbars 4a, 4b are insulated from each other by an intermediate insulation 4c and are additionally enclosed by an external insulation 4d, which increases the electrical safety and insulation integrity of the system.

[0070] System 20 establishes the electrical connection between busbars 4a, 4b of coaxial busbar 4, that are arranged coaxially and have their insulation stripped, and two external separate busbars 6a, 6b. System 20, together with coaxial busbar 4 and two separate busbars 6a, 6b, forms a connection assembly 1 that can be used as a DC connector within a charging inlet of an electric vehicle.

[0071] System 20 comprises a housing assembly 2, 3 that provides a receptacle 10 for coaxial

busbar **4**. This receptacle **10** extends in a direction of insertion E, which, in the embodiments illustrated, coincides with a longitudinal direction of extension of coaxial busbar **4**. Furthermore, housing assembly **2, 3** comprises a first receptacle **7a** for receiving one of separate busbars **6a** and a second receptacle **7b** for receiving other separate busbar **6b**, which enables precise positioning and connecting of busbars **6a, 6b**.

[0072] As can be seen in particular from the exploded views in FIGS. **1, 3, 5** and **7** and from the assembly sequence in FIGS. **10** to **12**, housing assembly **2, 3** is composed of a first housing half **2**, which comprises receptacle **10** for coaxial busbar **4**, and a second housing half **3**. Second housing half **3**, which is connected to first housing half **2** by being plugged together, contains both first receptacle **7a** for separate busbar **6a** as well as second receptacle **7b** for separate busbar **6b**.

[0073] First housing half **2** and second housing half **3** are made of plastic material, preferably glass fiber reinforced plastic material, to ensure high strength and durability. To ensure effective sealing after housing halves **2, 3** have been plugged together, a circumferential seal **30** is integrated between them.

[0074] After two housing halves **2, 3** have been plugged together, intermediate busbars **5a, 5b** are in an arrangement that enables a connection to busbars **4a, 4b** of coaxial busbar **4** in first housing half **2** and at the same time allows them to protrude into receptacles **7a, 7b** for separate busbars **6a, 6b** in second housing half **3**. Receptacles **7a, 7b** for separate busbars **6a, 6b** are configured such that, once housing halves **2, 3** have been plugged together, they are disposed at least in part in both housing halves **2, 3**. In addition, sections of receptacles **7a, 7b** can also be located exclusively within one of housing halves **2, 3**. This configuration enables efficient use of space and stable anchoring of separate busbars **6a, 6b**, as illustrated in the cross-sectional views according to FIGS. **2, 4, 6, 8, 9** and **10** to **12**.

[0075] The views in FIGS. **1, 3, 6** and **8** clearly show that first receptacle **7a** for one separate busbar **6a** and second receptacle **7b** for other separate busbar **6b** are integrated into housing assembly **2, 3** such that they have an orientation that is perpendicular to direction of insertion E. This enables a space-saving configuration of receptacles **7a, 7b** arranged parallel to one another and offset in a direction parallel to direction of insertion E of coaxial busbar **4**, which ensures optimal use of space within connection assembly **1**.

[0076] As illustrated by the perspective exploded views according to FIGS. **1, 3, 5** and **7** and the cross-sectional views parallel to direction of insertion E according to FIGS. **6** and **8**, system **20** comprises a first intermediate busbar **5a**, which is connected in a cohesive bond manner to one of busbars **4a** of coaxial busbar **4**, and a second intermediate busbar **5b**, which is likewise connected in a cohesive bond manner to other busbar **4b** of coaxial busbar **4**. Two intermediate busbars **5a, 5b** are each made of copper, which is crucial for their good conductivity and mechanical stability.

[0077] The cohesive bond connection between first intermediate busbar **5a** and second intermediate busbar **5b**, both made of copper, respectively to one of coaxially arranged busbars **4a, 4b** of coaxial busbar **4**, which are made of aluminum or an aluminum alloy, can be realized by various welding methods—including torsional ultrasonic welding, friction welding, electron beam welding, nanofoil welding or roll seam welding—or by soldering, such as induction brazing.

[0078] The cross-sectional views perpendicular and parallel to direction of insertion E, which in the present case coincide with a longitudinal direction of extension L of coaxial busbar **4**, as shown in FIGS. **2, 4, 6** and **8**, show that first intermediate busbar **5a** extends from receptacle **10** for coaxial busbar **4** to first receptacle **7a** for one of two separate busbars **6a**. Analogously, second intermediate busbar **5b** extends from receptacle **10** for coaxial busbar **4** to second receptacle **7b** for the other of two separate busbars **6b**.

[0079] As can be clearly seen in particular from FIGS. **1, 3** and **5** to **9**, two separate busbars **6a, 6b** can be arranged parallel adjacent to one another, more precisely offset parallel adjacent to one another in a direction that runs parallel to direction of insertion E. Accordingly, intermediate busbars **5a, 5b** can also be arranged parallel adjacent to one another, more precisely offset parallel

adjacent to one another in a direction that runs parallel to direction of insertion E. First intermediate busbar **5a** and second intermediate busbar **5b** are positioned within receptacle **10** in direction of insertion E such that they have a fixed distance from one another, which enables an orderly and efficient connection configuration.

[0080] To securely fasten intermediate busbars **5a**, **5b** to respective separate busbars **6a**, **6b**, screws **8a**, **8b** are used as touch-protected fastening elements that can be operated from outside housing assembly **2**, **3**. For this purpose, a first touch-protected screw **8a**, which is accessible from outside housing assembly **2**, **3**, is inserted through first intermediate busbar **5a** into corresponding receptacle **7a** for one busbar **6a**. Analogously, a second touch-protected screw **8b** is inserted through second intermediate busbar **5b** into receptacle **7b** for other busbar **6b**. This enables a robust and secure connection, as is particularly illustrated in the cross-sectional views according to FIGS. **6** and **8**.

[0081] First and second intermediate busbars **5a**, **5b** are fastened to separate busbars **6a**, **6b** by way of two screw connections using threaded sleeves **9a**, **9b**. First threaded sleeve **9a** is provided with an internal thread **24**, with which external thread **25** on screw end **23** of first screw **8a** engages, while second threaded sleeve **9b** comprises a corresponding internal thread **24** with which external thread **25** on screw end **23** of second screw **8b** engages. This form-fit connection ensures secure and permanent fastening, as can be seen from the cross-sectional views according to FIGS. **2**, **4**, **6**, **8** and **9**.

[0082] Screw shafts **22** of two screws **8a**, **8b** and threaded sleeves **9a**, **9b** are made of steel and comprise shoulders **19** with an enlarged diameter, each of which abuts a surface of intermediate busbar **5a**, **5b** facing away from respective busbar **6a**, **6b** of separate busbars **6a**, **6b**. This configuration ensures a reliable mechanical and electrical connection between intermediate busbars **5a**, **5b** and separate busbars **6a**, **6b** due to the screw connections, which is important for system integrity. The touch protection of screws **8a**, **8b** is obtained by the fact that their screw head **21** and oppositely disposed screw end **23** are each overmolded with plastic material.

[0083] As can be seen from the cross sections shown in FIG. **6** and FIG. **8**, each running parallel to direction of insertion E, touch-protected screws **8a**, **8b** are sized and positioned such that first screw **8a** ends in first receptacle **7a** for one separate busbar **6a**, while second screw **8b** ends in second receptacle **7b** for other separate busbar **6b**.

[0084] Threaded sleeves **9a**, **9b** made of steel and illustrated in FIGS. **1** to **4** and **9** to **12** fit precisely into openings **16** of separate busbars **6a**, **6b** and abut, with a radially projecting collar **27**, the surface of respective busbar **6a**, **6b** that faces away from respective intermediate busbar **5a**, **5b**. Tightening screws **8a**, **8b** in threaded sleeves **9a**, **9b** therefore ensures a firm mechanical and electrical connection between intermediate busbars **5a**, **5b** and separate busbars **6a**, **6b**. A variant illustrated in FIGS. **5** to **8** shows that separate busbars **6a**, **6b** can be constructed integrally with threaded sleeves **9a**, **9b**, which entails additional advantages in production and assembly.

[0085] There are openings **16** in separate busbars **6a**, **6b**, which in FIGS. **1** to **4** and **9** to **12** are configured as circumferentially closed round holes for receiving first threaded sleeve **9a** for first screw **8a** and second threaded sleeve **9b** for second screw **8b**. Alternatively, it is possible to configure these openings **16** as elongated holes to enable assembly flexibility or to configure them as round or elongated holes that are open on one side, which can facilitate alignment and fastening of the components during assembly.

[0086] In the illustrations of the first and third embodiments of system **20** according to the invention, as is evident in FIGS. **1**, **2**, **5** and **6**, it is characteristic for a cylindrical coaxial busbar **4**, its first busbar **4a** forming a cylindrical core and its second busbar **4b** forming a hollow cylindrical shell within coaxial busbar **4**, that first as well second intermediate busbar **5a**, **5b** comprise sleeve-shaped or hollow cylindrical sections as first connecting sections **11**, which are each arranged in receptacle **10** for coaxial busbar **4**.

[0087] These first connecting sections **11** of intermediate busbars **5a**, **5b**, in the form of sleeve-

shaped or hollow-cylindrical sections, are positioned in receptacle **10** for coaxial busbar **4** such that they engage at the circumference at least in part around exposed cylindrical inner busbar **4a** and exposed hollow-cylindrical outer busbar **4b** of coaxial busbar **4**. This configuration creates an extensive contact surface between intermediate busbar **5a**, **5b** and respective busbar **4a**, **4b** of coaxial busbar **4** in a cohesive bond connection, which can be realized by way of welding or soldering processes and thus ensures a high connection quality.

[0088] In the modifications shown in the second and fourth embodiment of system **20** according to the invention according to FIGS. **3**, **4**, **7** and **8**, coaxial busbar **4** has a respective flat, rectangular cross-sectional contour, where first busbar **4a** forms a flat, rectangular core in cross-section and second busbar **4b** forms a flat, rectangular and hollow shell in cross-section within coaxial busbar **4**.

[0089] Here, L-shaped angled elements are used as intermediate busbars **5a**, **5b**. A first leg **28a** of these L-shaped elements establishes a cohesive bond connection to corresponding flat, rectangular busbars **4a**, **4b** of coaxial busbar **4**. For this purpose, this first leg **28a** abuts, over its entire surface, a long underside of respective busbar **4a**, **4b** of coaxial busbar **4**. A second leg **28b**, which protrudes orthogonally from first leg **28a**, forms a planar section as second connecting section **12**, which is configured such that it comprises opening **14** for passing corresponding screw **8a**, **8b** through, thereby facilitating the assembly of the screw connections and enabling firm mechanical and electrical coupling.

[0090] The fifth embodiment shown in FIG. **9** shows intermediate busbars **5a**, **5b** in system **20** according to the invention which combine geometric elements of the preceding embodiments with one another. These intermediate busbars **5a**, **5b** have an arcuate connection **29a** between two legs **29b**, **29c** running parallel to one another. This structure nestles closely against the oppositely disposed long sides and a transverse side of flattened busbars **4a**, **4b** of coaxial busbar **4**, which creates an optimized contact surface for an effective connection. This configuration achieves an enlarged contact surface, which ensures a robust and reliable cohesive bond connection.

[0091] In the embodiments according to FIGS. **1** to **12**, screws **8a**, **8b** are each fastened in first housing half **2**. Each screw **8a**, **8b** is equipped with a touch-protected screw head **21** which is protected by plastic overmolding and comprises a plastic external thread **31**. This external thread **31** engages with a corresponding internal thread **32** of a fastening opening **17** in first housing half **2**. Fastening opening **17** in first housing half **2**, through which screws **8a**, **8b** are inserted, is covered at least in part by a screw holder **18a**, **18b** that is connected in a form-fit manner to this housing half **2**. This configuration secures screws **8a**, **8b** against accidental loosening and thus ensures a permanent and secure connection of housing halves **2**, **3**.

[0092] Openings **14** for the screw connections in fastening tabs **13** of intermediate busbars **5a**, **5b** are each configured as round holes closed all the way around, as shown in FIGS. **1**, **5** and **7**. In contrast, opening **14** in FIG. **3** is configured as an elongated hole open on one side. These versatile configuration options for openings **14**—whether as a round hole or an elongated hole, closed all the way around or open on one side—enable flexible adaptation to the given installation requirements and environments.

[0093] The optimized mechanical design of system **20** presented warrants reliability and safety. This includes robust mechanical stability, excellent electrical conductivity, effective touch protection and superior heat regulation, which are essential for long-term and safe use under a wide range of conditions.

[0094] The cross-sectional views according to FIG. **6** and FIG. **8**, which run along a sectional plane parallel to direction of insertion E, show partition walls **26**, which are arranged within housing assembly **2**, **3** between receptacles **7a**, **7b** for separate busbars **6a**, **6b** and which illustrate the spatial separation and structural support.

[0095] As shown in FIGS. **1**, **2** and **5** to **9**, first screw **8a** and second screw **8b** are each guided through a plastic sleeve **15a**, **15b**, which is inserted into an opening **14** of corresponding

intermediate busbar **5a**, **5b**. These plastic sleeves **15a**, **15b** extend from openings **14** to threaded sleeves **9a**, **9b**, thereby increasing touch protection. This design minimizes the risk of direct contact with the current-carrying parts and contributes to the safety of connection assembly **1**.

[0096] In accordance with the embodiments shown in FIGS. **3**, **4** and **10** to **12**, the use of plastic sleeves **15a**, **15b** can alternatively be omitted. In such configurations, intermediate busbars **5a**, **5b** are directly connected to corresponding busbars **6a**, **6b** of two separate busbars **6a**, **6b**. This direct connection simplifies the design and can increase reliability, even if it may not guarantee touch protection to the highest standards.

[0097] Depending on the configuration—with or without the integration of a plastic sleeve **15a**, **15b**—intermediate busbars **5a**, **5b** and associated separate busbars **6a**, **6b** can be spatially separated from one another, as shown in the embodiments according to FIGS. **1**, **2** and **5** to **9**. In such configurations, the screw connections establish not only a fixed mechanical connection, but also a direct electrical connection between intermediate busbars **5a**, **5b** and separate busbars **6a**, **6b**.

[0098] In FIG. **9**, the fifth embodiment, intermediate busbars **5a**, **5b** are configured as multi-layer metal sheets, which gives them a pronounced flexibility. This flexibility is ensured in particular along the plugging direction, i.e. parallel to the longitudinal axis of the screw. The illustration in FIG. **9**, highlighted by double arrows near arcuate connection **29a**, illustrates these flexibility properties of intermediate busbars **5a**, **5b**. In addition, limited movability of intermediate busbars **5a**, **5b** parallel to direction of insertion E is also made possible. As a result, these flexibility properties of intermediate busbars **5a**, **5b** improve the adaptability of connection assembly **1** to different assembly scenarios.

[0099] Although not explicitly shown in the present figures, a cooling device, in particular in the form of an integrated cooling channel, can be integrated into the structure of housing assembly **2**, **3** in order to ensure effective heat dissipation from all busbars **4a**, **4b**, **5a**, **5b**, **6a**, **6b** and thereby an optimized operating temperature during the charging process.

[0100] FIGS. **10** to **12** show an assembly sequence for producing a connection assembly **1** according to the invention.

[0101] According to FIG. **10**, first housing half **2** and second housing half **3** are each initially provided in a pre-assembled state. In this pre-assembled state, first housing half **2** is equipped with fastening elements **8a**, **8b** in the form of touch-protected screws already inserted into its fastening openings **17**, with coaxial busbar **4** inserted into its receptacle **10** in direction of insertion E, and with intermediate busbars **5a**, **5b**, which are connected in a cohesive bond manner to exposed busbars **4a**, **4b** of coaxial busbar **4**. Furthermore, second housing half **3** in this pre-assembled state is equipped with separate busbars **6a**, **6b** inserted into its receptacles **7a**, **7b**.

[0102] Screws **8a**, **8b** attached to first housing half **2** in FIG. **10** are not yet in a fully inserted pre-position. As symbolized in FIG. **10** by the thick arrow pointing vertically downwards, two pre-assembled housing halves **2**, **3** can therefore be plugged together to form a housing assembly **2**, **3**, where this plugging direction runs perpendicular to direction of insertion E.

[0103] FIG. **11** shows the end of the plugging process in which openings **14** in fastening tabs **13** of intermediate busbars **5a**, **5b** are aligned with openings **33** of threaded sleeves **9a**, **9b** fitted into openings **16** of separate busbars **6a**, **6b** in a direction perpendicular to direction of insertion E and to plugging direction according to FIG. **10**. In this intermediate assembly state, according to FIG. **11** with aligned openings **14**, **16**, **33**, inserting and tightening two screws **8a**, **8b** can now begin. The start of this insertion is symbolized in FIG. **11** by the two thick arrows at screws **8a**, **8b** in fastening openings **17** of first housing half **2**.

[0104] FIG. **12** shows the completion of the insertion and screw tightening process of screws **8a**, **8b**, whereby the assembly process is also completed by obtaining connection assembly **1** according to the invention. In the case of both screws **8a**, **8b**, each screw end **23** has been passed through opening **14** in fastening tab **13** and thereafter screwed with its external thread **25** into internal thread **24** of threaded sleeve **9a**, **9b**.

[0105] After completing this insertion and screw tightening process of two screws **8a**, **8b**, enlarged shoulder **19** on screw shaft **22** comes into contact with the surface of fastening tab **13** facing away from respective separate busbar **6a**, **6b**, as shown in FIG. **12**. Since threaded sleeve **9a**, **9b** also comprises a collar **27** enlarged in diameter that abuts a surface of respective separate busbar **6a**, **6b** that faces away from respective fastening tab **13**, tightening screws **8a**, **8b** creates a firm, force-fit connection between intermediate busbar **5a**, **5b** and separate busbar **6a**, **6b**. Since intermediate busbars **5a**, **5b** are again fastened in first housing half **2** and separate busbars **6a**, **6b** again in second housing half **3**, two housing halves **2**, **3** are then simultaneously firmly connected to each other by tightening screws **8a**, **8b**.

[0106] This results in extremely simple assembly. It should be emphasized that, in the embodiment according to FIGS. **10** to **12**, no separate plastic sleeves **15a**, **15b** are used like in the embodiments according to FIGS. **1**, **2** and **5** to **9**. Instead, the mutually facing surfaces of intermediate busbars **5a**, **5b** and separate busbars **6a**, **6b** come into direct pressing contact by simply inserting and tightening two screws **8a**, **8b**, which creates a secure mechanical and electrical connection in a simple manner between intermediate busbars **5a**, **5b** and separate busbars **6a**, **6b**.

[0107] Connection assembly **1** presented is characterized by numerous advantages that make it particularly suitable for applications in advanced electromobility and energy systems. The innovative combination of coaxial busbar **4** with intermediate busbars **5a**, **5b** not only optimizes electrical conductivity, but also contributes to reducing energy losses to a minimum. Due to its robust design and the implementation of touch-protected components, the assembly provides excellent mechanical stability and increased safety features. In addition, the designed simplicity of assembly enables quick and flexible installation, which can be easily adapted to different operating conditions and requirements.

[0108] While the invention has been described with reference to a preferred embodiment, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the spirit and scope of the invention as defined in the accompanying claims. One skilled in the art will appreciate that the invention may be used with many modifications of structure, arrangement, proportions, sizes, materials and components and otherwise used in the practice of the invention, which are particularly adapted to specific environments and operative requirements without departing from the principles of the present invention. The presently disclosed embodiments are therefore to be considered in all respects as illustrative and not restrictive, the scope of the invention being defined by the appended claims, and not limited to the foregoing description or embodiments.

Claims

1. A system for connecting two separate busbars to a coaxial busbar which comprises at least two busbars arranged coaxially to one another, with a housing assembly which comprises a receptacle for the coaxial busbar, wherein the receptacle extends in a direction of insertion and comprises a first receptacle for the one of the two separate busbars and a second receptacle for the other of the two separate busbars, with a first intermediate busbar for connecting the receptacle for the coaxial busbar to the first receptacle, wherein the first intermediate busbar is configured to be connectable to the one busbar of the coaxial busbar in an electrically conductive manner and with a second intermediate busbar for connecting the receptacle for the coaxial busbar to the second receptacle, wherein the second intermediate busbar is configured to be connectable to the other busbar of the coaxial busbar in an electrically conductive manner, and wherein the first and the second intermediate busbar can be arranged in the receptacle in the direction of insertion spaced apart from one another.

2. The system according to claim 1, wherein the current-conducting connection between the first intermediate busbar and the one busbar of the coaxial busbar and/or the current-conducting

connection between the second intermediate busbar and the other busbar of the coaxial busbar is or are configured as a cohesive bond connection.

3. The system according to claim 1, wherein with a first touch-protected fastening element that can be operated from outside the housing assembly and that projects through the first intermediate busbar into the first receptacle, and with a second touch-protected fastening element that can be operated from outside the housing assembly and that projects through the second intermediate busbar into the second receptacle.

4. The system according to claim 3, wherein at least one of the fastening elements is configured as a touch-protected screw.

5. The system according to claim 4, wherein the system comprises at least one threaded sleeve that is fastened to one of the two separate busbars, in particular inserted into an opening thereof, and is screwed to a touch-protected screw.

6. The system according to claim 1, wherein the first receptacle and the second receptacle in the housing assembly extend in a direction that runs transversely to the direction of insertion.

7. The system according to claim 1, wherein the first intermediate busbar and the second intermediate busbar each comprise a first connecting section that is arranged in the receptacle for the coaxial busbar and a second connecting section that is arranged either in the first receptacle for the one of the two separate busbars or in the second receptacle for the other of the two separate busbars.

8. The system according to claim 7, wherein the first connecting section is configured to be sleeve-shaped or hollow-cylindrical at least in part or planar at least in part.

9. The system according to claim 7, wherein a main direction of extension of the first connecting section is changed as compared to a main direction of extension of the first connecting section, and wherein this change in direction is formed in particular by an angular deformation between the connecting sections.

10. The system according to claim 7, wherein the second connecting section is configured as a fastening tab and this fastening tab comprises an opening through which the corresponding fastening element is guided.

11. The system according to claim 10, wherein the opening in the fastening tab is insulated from the corresponding busbar of the two separate busbars.

12. The system according to claim 1, wherein the first intermediate busbar and the second intermediate busbar are each configured as L-shaped elements.

13. The system according to claim 1, wherein the housing assembly comprises a first housing half in which the receptacle for the coaxial busbar is disposed, and a second housing half which can be plugged together with the first housing half and in which both the first receptacle as well as the second receptacle are disposed.

14. The system according to claim 13, wherein the intermediate busbars are arranged, at least in part, in the second housing half.

15. The system according to claim 3, wherein the fastening elements are attached to the first housing half.

16. A connection assembly comprising a system according to claim 1, a coaxial busbar as well as two separate busbars, wherein the one of the two separate busbars is connected to the first intermediate busbar by way of a first fastening element in the first receptacle and the other of the two separate busbars is connected to the second intermediate busbar by way of a second fastening element in the second receptacle.

17. The connection assembly according to claim 16, wherein fastening tabs of the intermediate busbars each directly abut a corresponding one of the two separate busbars.

18. The connection assembly according to claim 16, wherein, in the receptacle for the coaxial busbar, the intermediate busbars are each connected in a cohesive bond manner to a corresponding busbar of the coaxial busbar.

- 19.** The connection assembly according to claim 16, wherein a first housing half is configured to be pre-assembled with the fastening elements, the coaxial busbar, and the intermediate busbars, and a second housing half is configured such that it can be plugged together with the first housing half.
- 20.** The connection assembly according to claim 19, wherein the second housing half is configured to be pre-assembled with the two separate busbars.
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